Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (MECHANICAL/IP ENGINEERING)

END SEMESTER EXAMINATIONS, NOV 2019

SUBJECT: LEAN MANUFACTURING [MME 4007] (PE VI)

REVISED CREDIT SYSTEM
(28/11/2019)

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **all** questions.
- Missing data may be suitably assumed.

1A. Write a short note on Poka Yoke. **(3)**

1B. Explain with an example, the importance of selecting a product family in value stream mapping. **(3)**

1C. Explain the job description for a value stream manager. **(4)**

2A. Explain 'paced withdrawal' with a neat sketch. **(3)**

2B. Calculate the time in each shift for improvement in a company for a month by considering 30 days in a month, 4 holidays, 8 hours per shift, two shifts per day, 60 minutes for monthly safety meeting, 60 minutes for weekly kaizen meeting and 120 minutes for TPM, PM and 5s cleaning. **(3)**

2C. Explain the points to be considered to design a Kanban. **(4)**

3A. What is economic order quantity? Explain with a numerical example. **(3)**

3B. With a flow chart explain the framework of Toyota's production smoothing. **(3)**

3C. Explain how to maintain production smoothing during fluctuating market demand. **(4)**

4A. Differentiate between MRP and Kanban. **(4)**

4B. Write a short note on calculating buffer in Kanban. **(3)**

4C. Draw organization structure for promoting setup time reduction **(3)**

5A. Differentiate product flow layout and job shop layout. **(3)**

5B. Explain shortening processing time through small-sized lot production **(3)**

5C. In value stream mapping justify the statement – value stream mapping ties together lean concepts and techniques, which helps you avoid “cherry picking”. **(2)**

5D. In shortening setup time justify the statement – although mechanical systems are very convenient, an expensive investment in them would be “putting the cart before the horse” **(2)**

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